Transmission of signals

The message may be a function $f(x, y, t)$ of two space coordinates and time, the light intensity at point (x, y) and time t on a pickup tube plate. Two or more functions of time, say $f(t)$, $g(t)$, $h(t)$ — this is the case in three dimensional sound transmission or if the system is intended to service several individual channels in multiplex. The logarithmic measure is more convenient: $\log_2 M = \log_{10} M / \log_{10} 2 = 3.32 \log_{10} M$, and a device with two stable positions, such as a relay or a flip-flop circuit, can store one bit of information. N such devices can store N bits, since the total number of possible states is 2^N and $\log_2 2^N = N$. If the source has n symbols with probabilities $(p_1, p_2, \ldots, p_n)$ then the entropy $H = -\sum_i p_i \log p_i$ measures, in bits per symbol, the information produced. Several functions (u, v, w) of several variables (x, y, z, t) arise in colour television, and the channel capacity $C = \lim_{T \to \infty} (\log N(T))/T$ bounds the rate of transmission.